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Polymer tantalum capacitors and lifetime

**[JUNE 23, 2023 <](https://www.edn.com/polymer-tantalum-capacitors-and-lifetime/)****[HTTPS://WWW.EDN.COM/POLYMER-TANTALUM-CAPACITORS-AND-LIFETIME/>](https://www.edn.com/polymer-tantalum-capacitors-and-lifetime/)****[BY JOHN DUNN <](https://www.edn.com/author/john-dunn/)****[HTTPS://WWW.EDN.COM/AUTHOR/JOHN-DUNN/>](https://www.edn.com/author/john-dunn/)****[COMMENTS 6 <](https://www.edn.com/polymer-tantalum-capacitors-and-lifetime/#comments)****[HTTPS://WWW.EDN.COM/POLYMER-TANTALUM-CAPACITORS-AND-LIFETIME/#COMMENTS>](https://www.edn.com/polymer-tantalum-capacitors-and-lifetime/#comments)**

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The subject of capacitors seems quite attractive from the standpoint of having low values of equivalent series resistance (low ESR) but there might be an issue which we see when we extract a small portion of a chart about those capacitors as taken from the Digi-Key website where we find ratings for “Lifetime @ Temp.”. Please see the following (**Figure 1**) excerpt from **[Digi-Key <](https://www.digikey.com/products/en/capacitors/tantalum-polymer-capacitors/70?k=polymer+tantalum+capacitors&k=&pkeyword=polymer+tantalum+capacitors&sv=0&p)**

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:

Manufacturer	Series	Capacitance	Operating Temperature	Lifetime @ Temp.
AVX Corporation	-	0.47µF	-55°C ~ 85°C	
KEMET	F38	0.68µF	-55°C ~ 105°C	1000 Hrs @ 105°C
Panasonic Electronic Components	KO-CAP® T520	1µF	-55°C ~ 125°C	1000 Hrs @ 125°C
Rohm Semiconductor	KO-CAP® T521	1.5µF	-55°C ~ 150°C	1000 Hrs @ 85°C
Vishay Sprague	KO-CAP® T522	2.2µF	-55°C ~ 85°C	2000 Hrs @ 105°C
	KO-CAP® T523	3.3µF		2000 Hrs @ 125°C
	KO-CAP® T525	3.9µF		2000 Hrs @ 85°C
	KO-CAP® T527	4.7µF		
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Figure 1 Polymer tantalum capacitor lifetime at temperature (right). Source: Digi-Key

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In all these years, I had never before noticed a “lifetime” rating for any other capacitor types. For me, this specification raised heretofore unknown concerns.

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From a different URL which unfortunately is no longer active, I found a Q & A as shown in Figure 2

<https://www.vishay.com/docs/42106/faqpolymertantalumcaps.pdf> <
<https://www.vishay.com/docs/42106/faqpolymertantalumcaps.pdf>>. We would

Q:

Do polymer tantalum capacitors have issues with out-gassing?

A:

Polymer tantalum capacitors are thoroughly dried and vacuum sealed in aluminum foil bags so there will be no out-gassing when the parts are used properly. However, polymer tantalum caps are rated Moisture Sensitivity Level 3 (MSL 3) and they will slowly absorb moisture once unsealed and exposed to the environment. Out-gassing may occur during reflow mounting if the parts are handled in violation of MSL 3 required practices.

Figure 2 An application note excerpt from Vishay for polymer tantalum capacitors.

Source: Vishay

Please take special note of the words “....slowly absorb moisture...”.

I downloaded the Vishay-Sprague file “[polymerguide < https://www.edn.com/wp-content/uploads/polymerguide.pdf>](https://www.edn.com/wp-content/uploads/polymerguide.pdf)” which seems to describe these capacitors quite well, but I found no information in that document regarding “lifetime”. Thus, the question in my mind is whether the anticipated “lifetime” of polymer tantalum capacitors is the result of aging brought about by slow but inexorable absorption of moisture from the environment.

Would a circuit board coating (parylene or something similar) quell that moisture absorption aging process and extend the capacitors’ expected lifetime?

I do not know but *SOMETHING* is limiting “lifetime”. Taking whatever it is to be an aging process opens up a plausible line of reasoning via Arrhenius’s Law which briefly stated is that any aging process takes place at a rate which doubles for each ten degrees Celsius rise in temperature. Conversely, the rate of an aging process gets cut in half for each ten degrees Celsius fall in temperature.

This aging issue has been addressed before regarding [film resistors < http://licn.typepad.com/my_weblog/2010/12/resistor-aging-rates-vs-temperature-john-dunn-consultant-ambertec-pe-pc.html>](http://licn.typepad.com/my_weblog/2010/12/resistor-aging-rates-vs-temperature-john-dunn-consultant-ambertec-pe-pc.html).

Applying Arrhenius’ Law to this capacitor situation, we can estimate as follows:

$$\text{Lifetime} = (\text{Lifetime at Max. Temperature}) * 2^{((\text{Max. Temperature} - \text{Degrees}) / 10)}$$

The variable “degrees” is the actual temperature to which the capacitors will be exposed.

For the six cases shown on the Digi-Key chart in Figure 1, we extrapolate six “lifetime” versus temperature curves in the plots of **Figure 3**. The little dots along each curve are where “lifetime hours” gets doubled for each 10°C drop in temperature from their respective maximums.

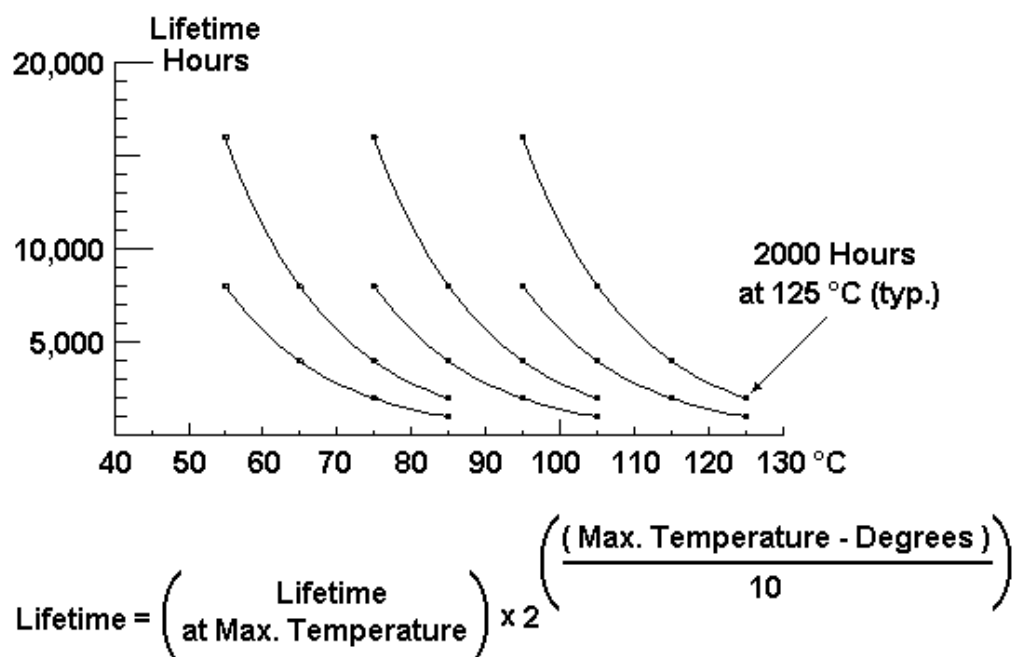


Figure 3 Extrapolated lifetimes versus temperature, lifetime hours are doubled for every 10⁰C drop in temperature. Source: John Dunn

John Dunn < <https://www.edn.com/author/john-dunn/> > is an electronics consultant, and a graduate of The Polytechnic Institute of Brooklyn (BSEE) and of New York University (MSEE).

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6 COMMENTS ON “POLYMER TANTALUM CAPACITORS AND LIFETIME”



John Dunn

June 23, 2023

This comment by Tom came directly to me. For some reason, he wasn't able to access this page.

From: Thomas Look [mailto:messages-noreply@linkedin.com]

Sent: Friday, June 23, 2023 7:11 PM

To: John Dunn

Subject: Re: Polymer Tantalum Capacitor Lifetime

John Dunn

Hi John,

For some reason, I am unable to register an account on EDN, but I wanted to comment on your article regarding Polymer Tantalum Capacitor Lifetime.

I heard some Kemet engineers mention on a podcast that the lifetime for Polymer Aluminum Capacitors increases by x10 for every 20 deg C below the rated lifetime temp. I found this note which mentions the same thing:

<https://www.kemet.com/en/us/technical-resources/high-vibration-applications-with-kemets-solid-polymer-aluminum-capacitors.html> <
<https://www.kemet.com/en/us/technical-resources/high-vibration-applications-with-kemets-solid-polymer-aluminum-capacitors.html>>

I don't know if that applies to Polymer Tantalum Capacitors as well. However, here are two notes from Kemet with info on Polymer Tantalum lifetimes:

<https://www.kemet.com/en/us/technical-resources/polymer-tantalum-capacitors-continue-to-toughen-up-to-meet-automotive-demands.html> <
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Best,

Tom

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**Konstantin Kim**

June 26, 2023

Interesting!

FYI, archived link to “absorb moisture” FAQ:

<https://web.archive.org/web/20190713170118/https://www.vishay.com/docs/421>

<

<https://web.archive.org/web/20190713170118/https://www.vishay.com/docs/421>[↪ Log in to Reply](#)**SteveHageman**

June 30, 2023

I too used to think that you can “Calculate your way to reliability” having been exposed to Mil-HDBK-217, BELCORE, etc. early in my professional career.

Then in the early 1990’s I met a fellow by the name of: Charles Leonard at a POWERCON meeting. There Mr. Leonard presented a paper that laid bare the assumption that the equations were even applicable or right.

I also recently found one his papers from 1990 and have put it on the internet archive at,

<https://archive.org/details/charles-leonard-cost-effective-design-technques-for-electronics-relaibility-1990> <<https://archive.org/details/charles-leonard-cost-effective-design-technques-for-electronics-relaibility-1990> >

Quoting from Mr. Leonard,

“Calculated results are routinely used as though they present an accurate model of the temperature-related reliability of devices. They do not.”

I also recommend looking at Kenneth Watts article: “A Unifying Approach to Designing for Reliability” in the July 1981 issue of the HP Journal,

https://archive.org/details/hp_journal_1981-07/page/n23/mode/2up <https://archive.org/details/hp_journal_1981-07/page/n23/mode/2up >

Cheers.

[↪ Log in to Reply](#)**SteveHageman**

June 30, 2023

I also wrote about Mr. Leonard’s 1990’s POWERCON talk in 2012 as per the link below (I had trouble finding it at first), it’s too bad that this was pre-internet as his talk would have made an very useful training video that is applicable even today.

<https://www.edn.com/a-trip-through-quality-part-3/> <<https://www.edn.com/a-trip-through-quality-part-3/> >[↪ Log in to Reply](#)

**John Dunn**

June 30, 2023

My dirty little secret is that in all the years I designed this or that, I never assumed that any of the components I used, except for vacuum tubes, would have a limited service lifetime. I had working "All American Five" AC/DC radios at home that were older than I was. Component lifetimes were never seen as an issue. Arrhenius' Law, moisture absorption, none of that factored in.

Derating guidelines per this or that standard were adhered to because customers insisted on it. Every voltage level and every current level and every power level could not exceed some fixed percentage of a device's published rating(s) or the customer would not accept delivery and the company would not get paid.

Each case needed to be analytically proven too. I sometimes felt that a contractual standard was questionable, but compliance was mandatory.

Everything worked.

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imreac

July 18, 2023

Capacitor Design Rules:

A: Use Capacitors with the highest voltage and temperature rating you can fit (in the product and budget). This also applies to MLCCs.

B: Use tantalums only IF 1) you know for sure the rated voltage will not be exceeded, even for milliseconds, under all operating and surge conditions, 2) the assemblies are 100% inspected for proper polarity installation, and 3) the government is paying for them.

C: Keep them away from heat! Don't prop, glue or heat shrink a TO-220 transistor or rectifier to a capacitor (or even close). Same for transformers, inductors and power resistors. Get a \$100 IR "gun" and check the temps on your prototype in the highest power dissipation mode. Move the capacitors as far away from the hot stuff as possible.

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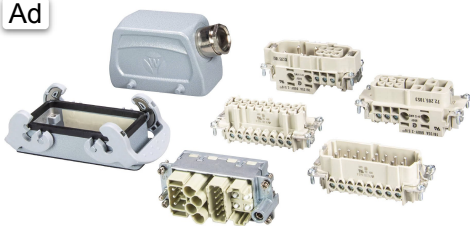
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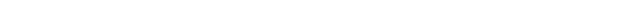
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